Yan Liu

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Education and Training

September 2026 — Present: Shanghai Jiao Tong University

Postdoctoral Fellow, Mechanical Engineering; Mentor: Zhen Huang (CAE Member, Chair Professor)

September 2021 — March 2026: Shanghai Jiao Tong University

Ph.D. Nuclear Science, M.S. Nuclear Science; Advisor: Xiaojing Liu (Distinguished Professor)

September 2017 — June 2021: Shanghai Jiao Tong University

B.Eng. Nuclear Engineering; Advisor: Tengfei Zhang (Professor)

Selected Honors and Awards

- First Prize of the China Electric Power Science and Technology Progress Award, 2026
- Best Paper Award of International Conference on Nuclear Engineering (ICONE), 2024
- European Nuclear Education Network (ENEN) ENEN++ Nuclear Mobility Fund, 2024

Research — Summary

- Overview: My research focuses on interfacial heat and mass transfer, multiphase transport, and the coupled dynamics of fluids, particles, and materials under complex operating conditions. My work explores interfacial transport, particle deposition, and physicochemical interactions in multiphase and porous systems to address challenges in energy conversion, thermal management, and system reliability. My research integrates advanced experimental characterization with theoretical and computational modeling, drawing on principles of fluid mechanics, heat and mass transfer, colloidal and interfacial science, materials science, and multi-physics coupling. Notable contributions include:
  - Interfacial Transport and Phase-Change Phenomena —
    - Elucidated the coupled mechanisms of heat and solute transport near phase-change interfaces, revealing their roles in local concentration, precipitation, and deposition.
    - Investigated the interplay between capillary transport and pore evolution in hierarchical porous structures, demonstrating its influence on salt-tolerant boiling performance.
  - Multiphysics Transport in Porous Media —
    - Developed coupled models integrating fluid flow, heat transfer, solute migration, and porous-structure evolution to characterize nonequilibrium transport and deposition.
    - Quantified the cross-scale effects of localized deposition and solute accumulation on macroscopic thermal performance and system-level responses.
  - Experimental Characterization and Surface Engineering —
    - Developed operando experimental approaches for reproducing and characterizing particle deposition and material evolution under controlled operating conditions.
    - Integrated surface characterization, experimental measurements, and computational analysis to investigate physicochemical interactions and interfacial processes.

Publications

* denotes corresponding author, † denotes equal contribution

Journal Articles

18. Critical heat flux on boiling induced self-assembly surfaces
    Chuandong Liu, **Yan Liu**, Hui He*, Tengfei Zhang & Xiaojing Liu*
    International Journal of Heat and Mass Transfer 257, 128230 (2026)
    Relevant research areas: Functional and Smart Surfaces, Energy Conversion and Utilization

17. Heat transfer transition of fuel rod caused by CRUD growth
    Guolian Wang, **Yan Liu**, Xiaojing Liu*, Xiang Chai, Tengfei Zhang & Hui He*
    International Journal of Thermal Science 220A, 110268 (2026)
    Relevant research areas: Energy Conversion and Utilization

16. Experimental study on the influence of epoxy patterns of bi-conductive surfaces on pool boiling
    Longchang Tang, Wei Xu, **Yan Liu** & Xiaojing Liu*
    Nuclear Engineering and Design 459, 115178 (2026)
    Relevant research areas: Functional and Smart Surfaces, Energy Conversion and Utilization

15. Pressurized water reactor fuel corrosion-related unidentified deposit and its related safety issues – II. Corrosion product deposition and heat transfer modeling
    Yan Liu, Jiajie Chen, Guolian Wang, Hui He*, Tengfei Zhang & Xiaojing Liu*
    Annals of Nuclear Energy 211, 110932 (2025)
    Relevant research areas: Interfacial Transport

14. Review on corrosion-related unidentified deposit of pressurized water reactors
    Hui He*, **Yan Liu**, Shiwei Wang, Tengfei Zhang, Xiang Chai & Xiaojing Liu*
    Nuclear Engineering and Design 441, 114095 (2025)
    Relevant research areas: Interfacial Transport

13. High-resolution multi-physics analysis of CRUD effects in reactor assemblies
    Chaoyuan Zhang, Han Yin, **Yan Liu**, Hui He, Xiaojing Liu, Tengfei Zhang* & Sicheng Wang
    PRX Energy 4(3), 033016 (2025)
    Relevant research areas: Energy Conversion and Utilization

12. Understanding flow fouling deposition and solute hideout–return behavior at the phase change interface
    Yan Liu, Xiaojing Liu*, Tengfei Zhang & Hui He
    ACS Applied Materials & Interfaces 16(4), 4719–4728 (2024)
    Relevant research areas: Interfacial Transport

11. Synthesizing nuclear power plant fouling with fractal characteristics enables an in-depth study of concerned nuclear safety issues
    Yan Liu, Xiaojing Liu*, Hui He, Tengfei Zhang & Xiang Chai
    iScience 27(2), 108789 (2024)
    Relevant research areas: Functional and Smart Surfaces

10. Pressurized water reactor fuel corrosion-related unidentified deposit and its related safety issues – I. Corrosion product deposition and heat transfer
    Yan Liu, Hui He*, Tengfei Zhang & Xiaojing Liu*
    Annals of Nuclear Energy 208, 110758 (2024)
    Relevant research areas: Interfacial Transport

9. Pressurized water reactor fuel corrosion-related unidentified deposit and its related safety issues – III. N-TH-M coupled CIPS prediction
   Yan Liu[†], Guolian Wang[†], Hui He[*], Tengfei Zhang & Xiaojing Liu[*]
   Annals of Nuclear Energy 208, 110800 (2024)
   Relevant research areas: Energy Conversion and Utilization

8. Study on coupling effect of CRUD growth and heat transfer on fuel rod
   Guolian Wang, Yan Liu, Xiaojing Liu[*] & Hui He[*]
   International Journal of Thermal Science 199, 108892 (2024)
   Relevant research areas: Interfacial Transport, Energy Conversion and Utilization

7. Thermodynamics and kinetics of zinc addition in pressurized water reactor primary circuit
   Yan Liu, Xiaojing Liu[*], Tengfei Zhang & Hui He
   Yuanzineng Kexue Jishu/Atomic Energy Science and Technology 58, 961–969 (2024)
   Cover Article
   Relevant research areas: Chemical Process Dynamics

6. Research on reproduction method of CRUD depositions based on anodic oxidation
   Yan Liu, Xiaojing Liu[*], Sijia Du, Jiageng Wang & Hui He
   Hedongli Gongcheng/Nuclear Power Engineering S1, 46-50 (2023)
   Relevant research areas: Functional and Smart Surfaces

5. Two-phase boiling heat and mass transfer characteristic within PWR CRUD deposition
   Yan Liu, Xiaojing Liu[*] & Hui He[*]
   Yuanzineng Kexue Jishu/Atomic Energy Science and Technology 56, 1551-1558 (2022)
   Relevant research areas: Interfacial Transport

Submitted / Under Review

4. Coupled interfacial transport enables salt-tolerant boiling in hierarchical porous surfaces
   Yan Liu, Xinyu Li, Chuandong Liu, Hui He[*], Tengfei Zhang & Xiaojing Liu[*]
   Relevant research areas: Interfacial Transport, Functional and Smart Surfaces

3. ICBA: High-fidelity multi-physics model for CRUD boron hideout in pressurized water reactors
   Yan Liu, Tengfei Zhang, Hui He[*] & Xiaojing Liu[*]
   Yuanzineng Kexue Jishu/Atomic Energy Science and Technology 56, 1551-1558 (2022)
   Relevant research areas: Interfacial Transport, Energy Conversion and Utilization

2. Accelerating global low-carbon energy transitions with regionally optimized nuclear–renewable microgrids
   Xinyu Li, Yan Liu, Hongtao Bi, Xiang Chai[*] & Xiaojing Liu[*]
   Relevant research areas: Smart Grid

1. In-situ cold neutron characterization of subtle coupled signatures in background-obscured microscale deposits
   Yixuan Li, Guangchao Yang, Xin Yang, Hui He[*], Tengfei Zhang, Shiwei Wang, Zhengyang Gao, Yan Liu, Rundong Li, Heyong Huo, Sheng Wang & Xiaojing Liu[*]
   Relevant research areas: Advanced Characterization

Conference Papers

5. Dynamic simulation and performance analysis of a nuclear-solar energy system with thermal energy storage
   Xinyu Li, Yan Liu, Xiang Chai & Xiaojing Liu*
   The 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)

4. Experimental study on the influence of epoxy patterns of bi-conductive surfaces on pool boiling
   Longchang Tang, Wei Xu, Yan Liu & Xiaojing Liu*
   The 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)

3. High-fidelity CRUD coupling simulation for PWR power shift
   Chaoyuan Zhang, Han Yin, Yan Liu, Hui He, Xiaojing Liu & Tengfei Zhang*
   The 32nd International Conference on Nuclear Engineering (ICONE-32)

2. Study on thermal conductivity model for porous PWR CRUD depositions
   Yan Liu, Xiaojing Liu*, Hui He, Tengfei Zhang & Xiang Chai
   The 31st International Conference on Nuclear Engineering (ICONE-31)

1. Multi-physics coupling model for thermal hydraulics and solute transport in CRUD deposits
   Yan Liu, Xiaojing Liu*, Sijia Du, Jiageng Wang & Hui He
   The 23rd Pacific Basin Nuclear Conference (PBNC2022)

Honors and Awards

12. First Prize of the China Electric Power Science and Technology Progress Award, 2026
    Chinese Society for Electrical Engineering

11. Doctoral Graduate with Distinction: Graduate Excellence Award of Shanghai, 2026
    Shanghai Municipal Education Commission

10. Doctoral Student Program of the Young S&T Talents Cultivation Project, 2025
    The China Association for Science and Technology (CAST)

9. China National Scholarship (TOP 0.2%), 2025
    Ministry of Education of the People's Republic of China

8. Best Poster Award of the 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics, 2025
    American Nuclear Society (ANS), Korean Nuclear Society (KNS)

7. Best Poster Award of the 32nd International Conference on Nuclear Engineering, 2025
    The American Society of Mechanical Engineers (ASME), Chinese Nuclear Society (CNS)

6. Best Paper Award of the 31st International Conference on Nuclear Engineering, 2024
    The American Society of Mechanical Engineers (ASME)

5. ENEN++ Nuclear Mobility Fund, 2024
    European Nuclear Education Network (ENEN)

4. China National Scholarship (TOP 0.2%), 2024
    Ministry of Education of the People's Republic of China

3. President's Award, 2023
    Shanghai Jiao Tong University

2. Best Poster Award of the 23rd Pacific Basin Nuclear Conference, 2022
    Pacific Nuclear Council (PNC)

1. China National Scholarship (TOP 0.2%), 2022
    Ministry of Education of the People's Republic of China

References

- Prof. Xiaojing Liu (Ph.D. Advisor)
  Distinguished Professor in Energy, Shanghai Jiao Tong University
  Executive Vice Dean, College of Smart Energy, Shanghai Jiao Tong University
  Email: xiaojingliu@sjtu.edu.cn
- Prof. Xu Cheng
  Chair Professor in Applied Thermofluidics, Karlsruhe Institute of Technology
  Head, Institute for Applied Thermofluidics, Karlsruhe Institute of Technology
  Email: xu.cheng@kit.edu
- Prof. Tengfei Zhang
  Professor in Nuclear Science, Shanghai Jiao Tong University
  Member of the Leadership Team, Institute of Nuclear Science and Safety, Shanghai Jiao Tong University
  Email: zhangtengfei@sjtu.edu.cn